VILNIUS UNIVERSITY
ŠIAULIAI ACADEMY
BACHELOR PROGRAMME SOFTWARE ENGINEERING

Object-Oriented Programming assignment no 2

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- **Tasks:**

Create PHP a program to calculate the mathematical function 3x + 1 (Collatz conjecture)

- **Code:**

```
<?php

// Collatz Conjecture Program

echo "<h2>Collatz Conjecture Calculator</h2>";


// HTML form to take user input

echo '<form method="post">

Enter a number: <input type="number" name="num" required>

<input type="submit" value="Calculate">

</form>';


// Check if user submitted a number

if(isset($_POST['num'])) {

    $x = $_POST['num'];

    echo "<h3>Sequence:</h3>";

    while($x != 1) {

        echo $x . " → ";

        if($x % 2 == 0) {

            $x = $x / 2;

        } else {
```

```
        $x = 3 * $x + 1;
      }
    }
    echo "1";
}
?>
```

- **Output:**

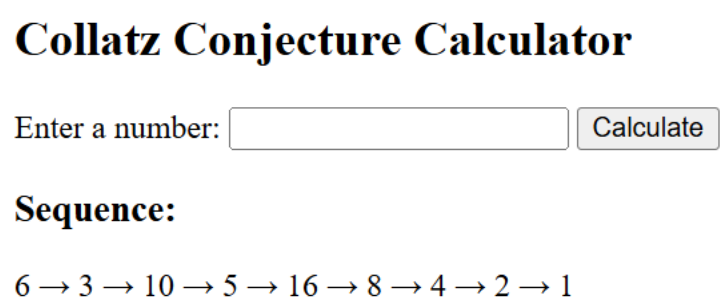

Figure 1: Input 6 and it gives output

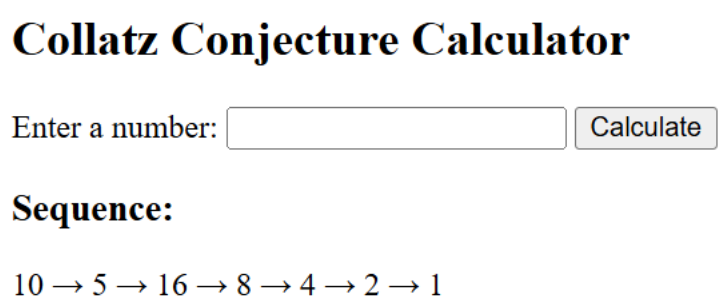

Figure 2: Input 10 and it gives output

- **Explanation:**

This PHP program calculates the Collatz Conjecture for any positive number. The user enters a number, and the program repeatedly applies the rules: divide by 2 if even, or multiply by 3 and add 1 if odd, until the number becomes 1. Each step of the sequence is displayed, showing how the number eventually reaches 1. For example, input 6 gives: 6 → 3 → 10 → 5 → 16 → 8 → 4 → 2 → 1.